

LLM - 11-Days Report.

Tokenization → Embeddings → Attention
→ FFN → Layer Norm → Activation

Transformers bu Hugging face
kutubxonasi xosoblanadi!

Architecture Variants

- Faster.
- Better.
- Less memory.
- Longer context.

Asosiy
maqsadlar

① Multi Head Attention = GPT 2 qurilish
Query Key Value o'zaro ishlaydi
Model n GPT 2 da 12 ta Head mavjud

② Multi Query Attention

③ Group Query Attention furlari

↓ GPT 2 da 12 ta ganyonali qurilmalar

Attention: MHA → GQA / MQA

Position: Absolute Embedding / ROPE

Activation: GELU / SWIGLU

Normalization: Layer Norm / RMS Norm

Vocabulary: BPE (50k) / Larger BPE / Sentence Piece
(GPT-4 da 256k token)

Goal: Good language modeling / Further Inference
context window

Residual - bu small hisoblanadi.
biri layerdan boshqa layerga
o'tayotganda ma'lumotlar tushib
qolishi mumkin. aynan shu
xolatlarini oldini olish uchun
residual ishlatiladi.

First residual

$$x_1 = x + \text{Attention}(\text{layer Norm}(x))$$

Second residual

$$x_2 = x_1 + \text{FFN}(\text{Layer Norm}(x_1))$$

Bu oyda Asosiy
2 ta nazariya

LLM ni ishlatish

LLM ni qurish

OLLama

Local AI tax joylashgan joy

App/play store = Local AI

Wrapper - No model loader, token
interface(MI).

Privacy, 0 cost

Yuklash yollari

~~use ollama~~ run llama3
~~use ollama~~ con